

A rule-based algorithm for tillage detection from Sentinel-1 time series

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1 Context

Tillage strongly affects **soil health**, influencing **C-dynamics** and **aggregate stability** [1-3]

Earth observation could support monitoring tillage practices, yet **optical indices lack reliability** due to cloud cover and are only sensitive to differences in crop residue cover

Can we use **Sentinel-1 SAR** data to estimate tillage dates and tillage type? Previous studies have shown promising results [4-6]

➔ Leveraging **coherence** (sensitive to **surface change)** and **backscatter** (sensitive to **soil roughness)**

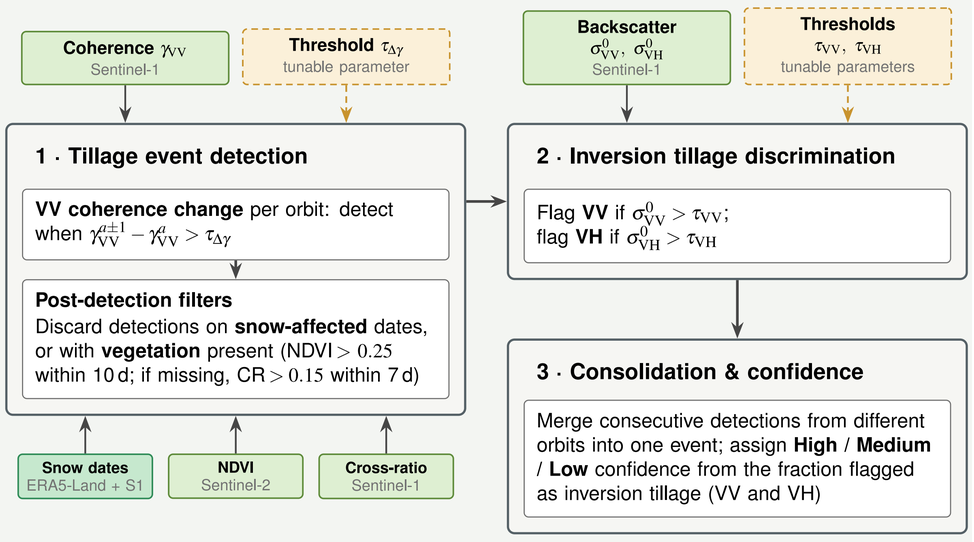
Terminology

Tillage: agricultural preparation of soil by mechanical agitation

Inversion tillage: The use of a mouldboard or disk plough to invert the topsoil, typically to a depth of 20-30 cm

2 Methods

Algorithm



Ground-truth data collection by weekly revisits

Site	Parcels	Observations	Tillage events	Inversion tillage events
Spring 2025	150	2127	167	29
Autumn 2025	106	1702	136	20

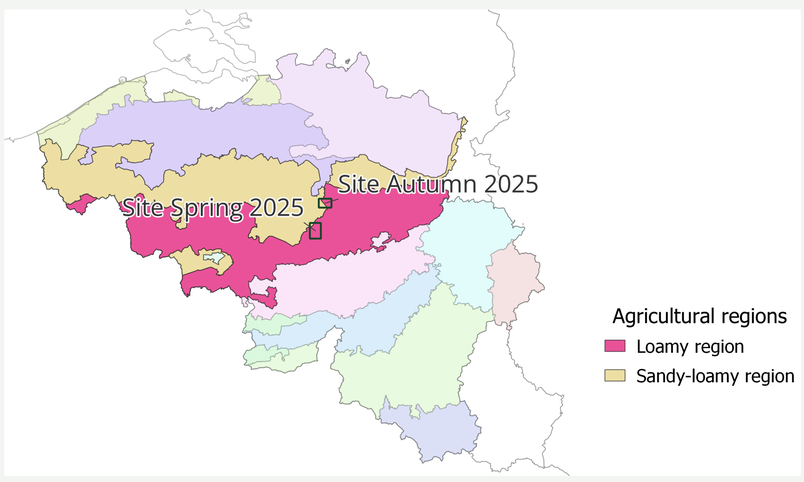


Figure 1. Location of the study sites in Belgium with respect to the official agricultural regions

3 Results

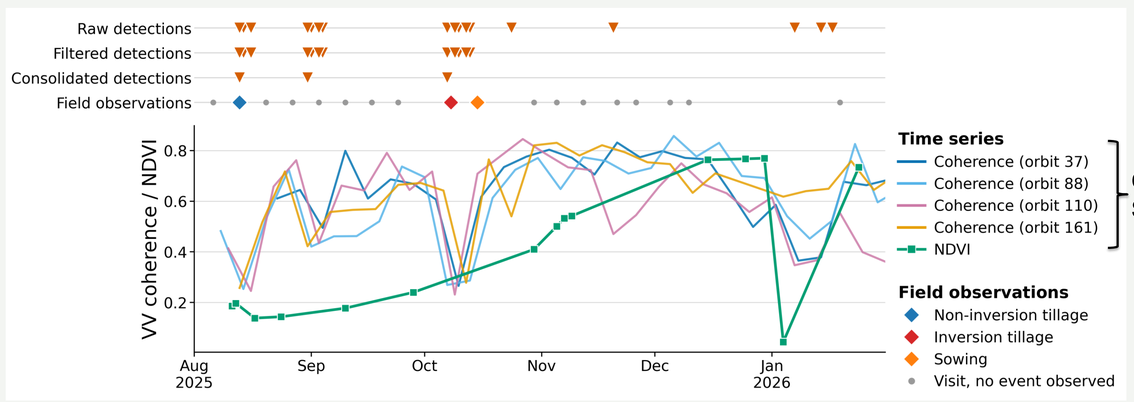


Figure 2. Example of the coherence change detections for a parcel with three recorded tillage events

STEP 1
Detect tillage events

0.25 fixed threshold on coherence change

0.48 → 0.69 **after filter**

Spring F1-score
without → with filters for snow and vegetation

STEP 2
Separate tillage types

	Spring 2025	Autumn 2025
VV	−10 dB	−8 dB
VH	−20 dB	−17 dB

Backscatter thresholds for inversion tillage

STEP 3
Consolidate & validate

F1-score against in-situ observations

	Spring 2025	Autumn 2025
Tillage detection	0.74	0.66
Inversion tillage	0.66	0.52

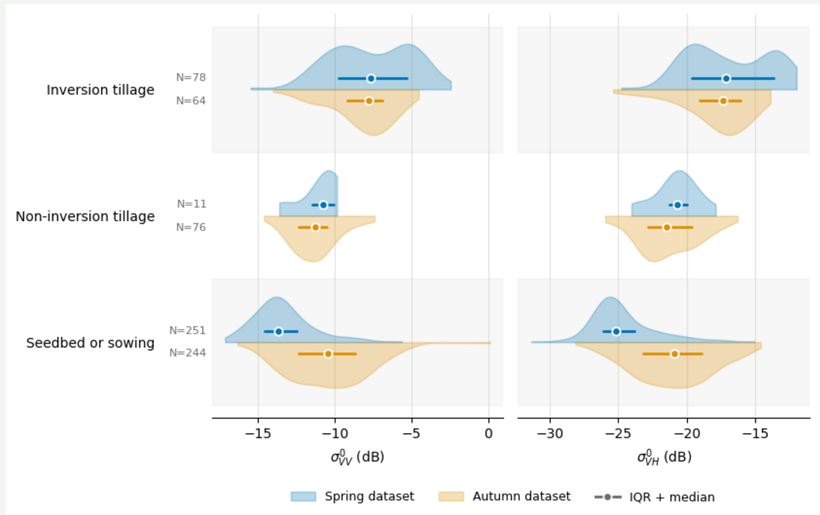


Figure 3. Distribution of backscatter values for three types of tillage on both study sites.

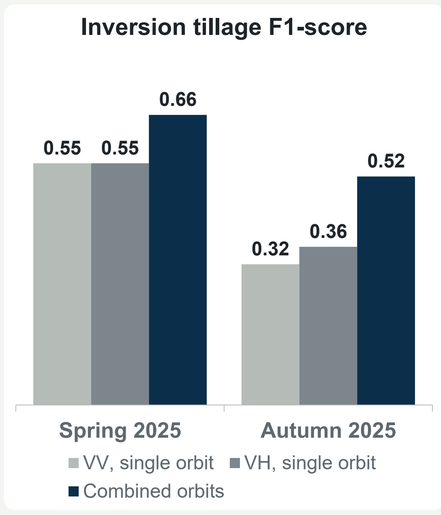


Figure 4. Combining overlapping orbits improves inversion tillage detection.

4 Conclusions

Coherence change can accurately **detect tillage events** with a threshold of 0.25 and applying filters for false detections

Types of tillage can be **differentiated by backscatter**, performance differs across seasons

Combining detections from **overlapping orbits increases accuracy**

References

[1] Clement, T. et al. Soil Use and Management 40, e13047 (2024).
[2] Meurer, K. H. E. et al. Earth-Science Reviews 177, 613–622 (2018).
[3] Six, J. et al. Soil Biology and Biochemistry 32, 2099–2103 (2000).
[4] Dingle Robertson, L. et al. Heliyon 9, e17322 (2023).
[5] Satalino, G. et al. IGARSS 2024, 1249–1252 (2024).
[6] McNairn, H. et al. IGARSS 2022, 5858–5861 (2022).

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Can you help evaluate our algorithm in other regions? Get in touch.

